Higher National Diploma in Engineering

First Year, Second Semester Examination - 2021

CE 1205 – Civil Engineering Construction I

QS-12044 – Civil Engineering Construction I

Instructions for Candidates:
Answer five questions only

No. of questions: 06
No. of pages: 03
Time: 03 hours

(01.)

I. Name the three stages in site investigations. (3 marks)

II. State four types of information which collect during the site investigations. (4 marks)

III. Write any four different activities carried out during site preparation for constructing a building. (4 marks)

IV. Describe problems associated with water in the subsoil. (3 marks)

V. Briefly explain following methods of subsoil water controlling.

a) Cofferdams b) Well point system (6 marks)

(02.)

I. State four design requirements of Ground floor construction. (4 marks)

II. Mention four types of floor construction methods. (2 marks)

III. What is the purpose of providing Skirting in a wall? (4 marks)

IV. Name four Clay brick-laying techniques with relevant sketches. (4 marks)

V. Write short notes for following shallow foundation types. You may use suitable sketches for explanation.

a) Pad foundation

b) Strip foundation

c) Mat foundation (6 marks)

(3.)

I. Write the advantages of a pitched roof compared to a flat roof of a building. (4 marks)

II. Briefly explain the importance of having heat insulation under a roof. Name two types of heat insulation techniques. (4 marks)

III. Well-designed Septic tank and soak pit can be used to minimize the environmental pollution. Justify this statement. (You may use suitable sketches if needed)(6 marks)

IV. State three causes of dampness in a building. Propose suitable construction techniques to mitigate/minimize the defects of dampness. (6 marks)

(4.)

I. Explain the difference between Putlog scaffoldings and Independent scaffoldings. (4 marks)

II. Name good safety practices in the usage of scaffolding. (4 marks)

III. State the functions of providing Openings in a building. (3 marks)

IV. What is the purpose of constructing a lintel above a door? (3 marks)

V. Formwork plays an important role in construction processes. Rationalize this statement with suitable examples with sketches. (6 marks)

(5.)

I. Briefly explain the function and the suitable building materials for following building components.

a) Ceiling

b) Columns

c) Partition walls (6 marks)

II. Write four objectives of plastering work in building construction. (4 marks)

III. Name four types of "building services" in a residential building. (4 marks)

IV. What is the difference between "Quality control" and "Quality assurance" in a building construction. Explain the importance of Quality assurance and control in this field. (6 marks)

(6.)

I. Explain the importance of a Contract agreement. (4 marks)

II. Explain the usage of the following construction machineries used in the building construction site

a) Dragline b) Pneumatic tired roller

c) Dredger d) Wheel trencher (4 marks)

III. Write four factors that should consider in selecting suitable construction equipment for a particular work at the site. (2 marks)

IV. Write 03 responsibilities of each of the following job roles in a construction site.

a) Site engineer

b) Planning engineer

c) Project manager

d) Structural engineer (8 marks)

V. Write the purpose of below mentioned documents used in the construction site.

a) Request for information

b) Daily report (2 marks)